

Per-Peptidoform Site-Determining-Ion Localization Score

Design proposal — the singleton regime for multi-peptidoform reporting

2026-07-06

1. Why we need this

We want to report **more than one peptidoform per peptide** (multiple positional isomers can be genuinely co-present), not just one max-weight winner per isomer group. The data already supports it: each isomer is its own `precursor_idx`, deconvolved per-scan and FDR'd independently, and `precursors_long` carries each peptidoform's best-PSM `iso_weight_fraction_at_scan`.

But the last experiment (10k-amol, 3 reps, 54,066 target-target vs 40,814 target-decoy peptidoforms after the `qval + global_qval` gate) showed a hard wall:

population	frac ≥ 0.75	frac ≥ 0.90	frac ≥ 0.95	floor
all peptidoforms	11.5%	10.7%	10.8%	~ 10.7% , never $< 10\%$
contested (≥ 2 competing siblings)	11.3%	8.2%	6.6%	~6%

The `iso_weight_fraction` cutoff cannot push per-peptidoform FLR below ~10%. The cause is diagnostic: at `frac  $\geq 0.75$` , ~half the confident calls are **singletons** — the peptidoform is alone in its isomer group at its apex scan (`iso_group_size_at_scan == 1`), so `frac = 1.0` *automatically* and carries **zero localization information**. Their target-decoy FLR (~11.6%) sets the floor.

A peptidoform is a singleton at its apex exactly when its sibling isomers elute at a **different RT** — which is the “genuinely co-present isomers” case we want to report. The sibling fraction measures **co-elution competition**; RT-separated isomers don't compete, so the fraction is blind to them. A peptidoform decoy that separates in RT with a clean (permuted) spectrum is indistinguishable from a real second form *by the fraction alone*.

Conclusion: the fraction is the right confidence for the co-eluting (contested) regime, but the RT-resolved (singleton) regime needs a different, spectral score. This document proposes that score.

2. Core idea

For a singleton peptidoform, we can't use co-elution competition, but we *can* ask the spectrum directly: **do the fragment ions that actually distinguish the assigned site from every alternative site support the assigned placement?** Those ions are the **site-determining**

ions. The score is a per-peptidoform, intrinsic measure — it needs only the observed spectrum at the peptidoform’s apex scan and the library fragments; it does **not** require the competing isomer to be detected.

Crucially, this is exactly the evidence that ID-level FDR does **not** protect. A loc-decoy passes `qual` mostly on **shared** ions (b/y ions whose cleavage does not span the moved site are identical between real and decoy). It should fail on the **site-determining** ions, because its mod mass sits on the wrong fragments. The site-determining score isolates precisely that discriminating evidence.

3. Definitions

3.1 Candidate sites

For a peptidoform of length L with a variable mod assigned to position a , the **candidate alternatives** are the other acceptor residues (for phospho: the other S/T/Y positions). Localization means: the mod is on a , not on any alternative a' .

3.2 Site-determining ions

General definition (competitors may differ at more than one site). A fragment distinguishes two same-composition isomers A and B iff their **cumulative modification content differs at that cleavage**: $(\# A\text{-mods at pos } \leq i) \neq (\# B\text{-mods } \leq i)$. Then A and B predict **different m/z** for it, offset by $(\Delta\text{count}) \times \text{mod_mass}$ — which may be $\pm 1\times, \pm 2\times, \dots$ the mod mass where mods pile up before a cleavage. Match each isomer at its **own** predicted m/z; the distinguishing-fragment set is per **competitor pair** and can be **non-contiguous and multi-magnitude** (e.g. $\{S_1, S_7\}$ vs $\{S_4, S_{10}\}$ distinguishes on $[1, 4)$ and $[7, 10)$ separately; $\{S_1, S_4\}$ vs $\{S_7, S_{10}\}$ gives a $+2\times$ shift on $[4, 7)$). So a distinguishing ion is **not** attributable to a single adjacent-site gap, and the competitor set is **all** same-composition isomers (not just single-site moves).

Single-site special case (the common one, used in the worked example below). When A and B differ at exactly one site — the mod is on a vs a' — the general test reduces to: the ion’s cleavage falls **between** a and a' (it contains exactly one of $\{a, a'\}$), so the mod mass ($+79.966$ for phospho) shifts the m/z for one placement but not the other.

- b_i (prefix, cleavage after residue i) contains positions $1..i$. It is site-determining for $\{a, a'\}$ iff exactly one of $a, a' \leq i$.
- y_j (suffix) contains positions $(L - j + 1)..L$. Site-determining iff exactly one of a, a' is in that range.

Worked example. Peptide ABCSDESFGK ($L = 10$), phospho candidate sites S_4 and S_7 .

```

position : 1  2  3  4  5  6  7  8  9 10
residue  : A  B  C  S  D  E  S  F  G  K
              ^cand a=4          ^cand a'=7
b-ions   : b1 b2 b3 b4 b5 b6 b7 b8 b9
y-ions   :      y9 y8 y7 y6 y5 y4 y3 y2 y1   (y_j covers L-j+1..L)

```

Ions whose cleavage lies in the gap $[4, 7)$ are site-determining:

- b_4, b_5, b_6 — contain S_4 but not S_7 -> carry $+80$ if the site is S_4 , unmodified if S_7 .
- y_4, y_5, y_6 — contain S_7 but not S_4 -> carry $+80$ if the site is S_7 , unmodified if S_4 .

So the real peptide (site S_4) shows $b_{4,5,6}$ at +80 and $y_{4,5,6}$ unmodified. A decoy claiming S_7 predicts the opposite and will **not** match the observed peaks on those six ions. The gap of distance $\delta = 3$ yields $\sim 2\delta = 6$ site-determining ions — matching the retention design already in `site_determining.jl`.

3.3 Phospho-specific caveat: neutral loss

pS/pT spectra are dominated by the **neutral loss of H3PO4** (−97.977) from the precursor and from mod-bearing fragments. Neutral-loss peaks are *not* site-determining (the loss can occur from any placement) and, worse, a b_i^{-98} can be isobaric with an unmodified b_i . **The scorer must exclude neutral-loss channels from the site-determining set** (or model them explicitly), or decoys will look better than they are. This is the single biggest correctness risk (see §9).

4. Score formulations

Let $I(\cdot)$ be the observed intensity of a matched peak (0 if unmatched within mass tolerance), computed at the peptidoform’s apex scan.

(A) SDI count — number of matched site-determining ions supporting the assigned site. Simple, but not abundance-normalized and ignores the alternatives’ evidence.

(B) SDI intensity fraction —

$$S_B = \frac{\sum_{\text{ion} \in \text{SDI, supports } a} I(\text{ion})}{\sum_{\text{ion} \in \text{SDI}} I(\text{ion at either placement})}$$

Normalizes for abundance; still pools all alternatives.

(C) Competitive site-determining ratio (recommended). Mirror the `iso_weight_fraction` logic but from spectral evidence. For each alternative a' :

$$s(a, a') = \frac{\text{support}(a)}{\text{support}(a) + \text{support}(a')}, \quad \text{support}(x) = \sum_{\text{ion} \in \text{SDI}(a, a')} I(\text{ion at placement } x)$$

The peptidoform score is the **worst case over all alternatives**:

$$S_C = \min_{a' \neq a} s(a, a')$$

i.e. “the assigned site must beat *every* alternative on its own discriminating ions.” $S_C \rightarrow 1$ when the site-determining ions cleanly support a ; $S_C \rightarrow 0.5$ when ambiguous; $S_C < 0.5$ when they actually favor an alternative.

Undefined case. If some alternative a' has **no observed site-determining ions** ($\text{support}(a) + \text{support}(a') = 0$), the site is **unlocalizable against a'** from this spectrum. This is a real, honest outcome (matches the literature: not every site is localizable). The scorer should return a distinct `n_sdi = 0` flag so we can *report the peptide but not the site*, rather than silently scoring it 0 or 1.

Recommendation: compute all three but calibrate on **(C)**, reporting **(A) n_sdi** alongside as the “is this even localizable” guard.

5. Two-regime localization confidence

The per-peptidoform localization decision becomes:

```
if iso_group_size_at_scan >= 2:      # contested / co-eluting
    confidence = iso_weight_fraction_at_scan    # existing; ~6.6% FLR at 0.95
else:                                # singleton / RT-resolved
    if n_sdi == 0: report peptide, flag site "unlocalizable"
    else:         confidence = S_C (site-determining ratio)
```

Both confidences are then thresholded to a **single target FLR**, calibrated jointly against the target-decoy population (§6). Longer term, (C) can *replace* the fraction everywhere (it is defined for contested peptidoforms too), but the two-regime split is the minimal first step and lets us keep the already-validated fraction where it works.

6. Calibration & FLR (unchanged framework)

FLR is estimated exactly as now — **target-target vs target-decoy among sequence-targets only, sequence decoys excluded** (they only set the upstream qval/global_qval). Every passing peptidoform, real or loc-decoy, gets the same S_C computed the same way:

$$\text{FLR}(\tau) = \frac{\#\{\text{target-decoy singletons} : S_C \geq \tau\}}{\#\{\text{target-target singletons} : S_C \geq \tau\}}$$

If S_C is a good localization discriminator, target-decoys — whose assigned site is wrong, so their site-determining ions won't match — collapse to low S_C , and FLR should fall well below the fraction's ~11.6% singleton floor. **The central empirical question this prototype answers: how far?**

7. What we can compute now vs. what needs extraction

Input	Source	Status
apex scan index per peptidoform	<code>new_best_scan</code> in <code>precursors_long</code>	[have] present
passing set (qval, global_qval)	<code>precursors_long</code>	[have] present
target-target / target-decoy label	<code>is_loc_decoy</code> join from library	[have] present
candidate sites, b/y positions	<code>site_determining.jl</code> (<code>acceptor_positions</code>)	[have] built
library fragment m/z + ion type + position + mod	library <code>fragments</code> table	[have] present
observed peaks (m/z, intensity) at apex scan	the run's Arrow MS2 spectra	[build] needs a read
fragment<->peak matching within tol	new targeted step	[build] to build

So this is **not** a pure `precursors_long` analysis — it is a **targeted extraction**: for each passing singleton, load its apex scan’s peaks from the run’s Arrow file (`new_best_scan`), match the library fragments, classify site-determining, compute S_C . It reuses the existing `site_determining.jl` machinery and runs offline on the current 10k results — **no re-search, no library rebuild**. Estimated scope: one focused extraction script (~150–250 lines) + reuse.

8. Validation plan

1. **Decoy calibration curve** — $\text{FLR}(\tau)$ vs τ on 10k singletons; compare to the ~11.6% fraction floor. Success = a threshold exists with $\text{FLR} \leq 5\%$ at usable coverage.
2. **Ground-truth check on standards** — the 225 Coon standards have known single sites. For the singleton subset of detected standards, compare S_C -thresholded **empirical** FLR (site vs truth) to the **decoy-estimated** FLR. Tests whether the decoy estimate is honest for singletons.
3. **Coverage cost** — IDs retained vs the fraction-only and winner-take-one baselines, at matched FLR.
4. **Unlocalizable rate** — fraction of singletons with `n_sdi` = 0 (report-peptide-not-site); this is a real property of the data, not a failure.

9. Risks & open questions

- **Neutral loss (biggest)**. Must exclude/model H_3PO_4 loss channels or decoys look too good. §3.3.
- **Shadow decoys may be too easy — again**. Decoy sites are non-acceptor “shadow” residues; their site-determining ions may be trivially discriminating, so S_C could **under-estimate** FLR just as the fraction did in the winner-take-one comparison. The **short-distance-biased** decoy variant is the proper calibrant; SDI scoring and better decoys are complementary, not alternatives.
- **Sparse site-determining ions**. Sites near a terminus, or with the discriminating ions below detection, give small `n_sdi`. Expect a non-trivial unlocalizable fraction — this is honest, but it caps how many singletons we can confidently report.
- **Fragment charge / isotopes / co-isolation**. Match must respect the fragment charge states in the library and avoid crediting a co-isolated precursor’s peak. Use the same tolerance the search used.
- **D/R normalization** (carried over): at ~1:1 target-decoy:target-target, D/R is the estimate; revisit vs $D/(R + D)$ if we change the decoy multiplicity.
- **Does S_C subsume the fraction?** Possibly — (C) is defined for contested peptidoforms too. If it matches or beats the fraction there, we unify on one score. To be tested, not assumed.

10. Proposed phased plan

- **Phase 0 (cheap, optional first look)**. From `precursors_long` alone: quantify the contested-vs-singleton split, how many peptides have ≥ 2 RT-separated real forms, and the target-decoy rate among singletons vs contested. Sizes the prize before building the extractor. (~1 script.)
- **Phase 1 (this proposal)**. Build the targeted SDI extractor; compute $S_C + \text{n_sdi}$ for all passing singletons at 10k; produce the decoy calibration curve (§8.1) and the standards

ground-truth check (§8.2). **Decision gate:** does a usable threshold reach $\leq 5\%$ FLR?

- **Phase 2 (if Phase 1 passes).** Wire the two-regime confidence (§5) into the reporting path; sweep to a target FLR; report multi-peptidoform IDs with per-site confidence and an `unlocalizable` flag.
- **Phase 3 (optional).** Replace the fraction with S_C everywhere if it dominates; and/or fold S_C , `n_sdi`, the fraction, and PSM score into a single trained localization model calibrated on the loc-decoys.

Companion docs: `site_determining_fragment_retention.md` (the ion-retention machinery this reuses), `FINDINGS.md` (the fraction-floor result motivating this), `isomer_decoy_shadow_sites.md` (the target-decoy construction).